Sims Framework Contract

Scope

This framework is fully independent from mut_var internals and writes outputs only under: - `sims/results/` - `sims/plots/`

It must remain compatible with mut_var inference by file contract.

Methods-doc readability requirement

Any methods/design Markdown in `sims/docs/` must also be rendered to PDF for human review.

Canonical command:

```
python sims/scripts/render_methods_pdf.py --docs-dir sims/docs --out-dir sims/docs/docs_pdf
```

This keeps generated human-readable artifacts within `sims/docs/`.

Canonical mut_var entrypoints

- Inference: `mutvar infer <observed.tsv> -o <infer.tsv>`
- Curve fit: `mutvar curve <infer.tsv> --fit-only -o <curve.tsv>`

No changes are made to `src/`, `tests/`, packaging, or dependencies.

Required observed schema

Observed TSV must include these exact columns: 1. `effect_allele_frequency` 2. `beta` 3. `standard_error`

Additional columns are allowed but optional.

Core simulation controls

The simulator must expose and record: - `frequency.demography_mode` (currently `equilibrium` implemented) - `effect.selection_mode` in `{hd, 1d}` - `ascertainment.ascertainment_statistic` with primary default `threshold_on_maf` and optional alternative modes `{threshold_on_v_true, threshold_on_v_hat}` - `ascertainment.maf_min` for MAF-threshold runs - `ascertainment.v_s_cutoff` and `ascertainment.p_value_threshold` for variance-threshold runs - `dfe.point_mass_zero` in `[0, 1)` with default `0.0`, defining the DFE atom at `beta_s = 0`; the tabulated SSD grid parameterizes the nonzero DFE component - `effect.trait_null_fraction` in `[0, 1]` with default `0.0`, defining an additional focal-trait null gate on top of latent DFE draws - fixed `n_ascertained` target per run - `truth_reference_n` for distribution-evaluation truth sampling (should be larger than `n_ascertained`)

Runtime caution: variance-threshold scenarios are intentionally retained for comparison checks but can be extremely sensitive to `v_s_cutoff` and may run for a long time if configured too strictly. For routine workloads, `threshold_on_maf` is the operational default in this framework.

Single-threshold contract for MAF ascertainment: when `ascertainment.ascertainment_statistic = threshold_on_maf`, generation support and ascertainment use the same threshold (`frequency.min_x` is overridden by `ascertainment.maf_min` for that run). For v-based ascertainment modes, generation support remains controlled by `frequency.min_x`.

The observed `beta` column is on the selection-scaled effect axis (`beta_s`), and metadata must record the canonical scaled coefficient definition `S_ud = 2Ne*s_ud`.

When `dfe.point_mass_zero > 0`, the DFE includes a nonzero atom at `beta_s = 0` in latent effects before noise and ascertainment. Observed outputs are not required to preserve an exact zero fraction.

When effect.trait_null_fraction > 0, additional loci can have latent beta_s=0 even when selected/neutral draws remain nonzero under the DFE. Observed outputs are not required to preserve an exact zero fraction.

Domain and quality constraints

For every row in observed TSV: - effect_allele_frequency is finite and in [0, 1] - beta is finite - standard_error is finite and strictly > 0 - No nulls in required columns - File is tab-separated with a header row

Simulator output artifacts per run

Given run_id, simulator writes: - sims/results/<run_id>.observed.tsv (mut_var input) - sims/results/<run_id>.truth.tsv (latent variables and generating params, including zero-source flags; may use an independent larger truth-reference sample) - sims/results/<run_id>.meta.tsv (run metadata and summary stats) - sims/results/<run_id>.manifest.json (reproducibility metadata)

Downstream mut_var outputs: - sims/results/<run_id>.infer.tsv - sims/results/<run_id>.curve.tsv

Recovery objective

Primary study question: - How well mut_var's inferred mixture captures the underlying DFE-generated effect-size distribution.

Evaluation compares: - Truth-side unconditional effect distribution summaries - mut_var inferred mixture summaries reconstructed from infer.tsv (mu0, var0, and mixture weights) - Distribution distances and tail-fidelity metrics on $|\beta_s|$

AF-conditioned diagnostics may be computed as optional secondary outputs, but they are not the primary criterion in this workflow.

Reproducibility requirements

Each run must be reproducible from: - random seed - scenario configuration - exact commands executed - artifact paths and basic checks (row counts and key summary values)